Electrical BUS Literature Review

Overview of the Electrical BUS Subsystem

The Electrical BUS subsystem's goal is to provide the CubeSat with a centralized point of control, which at a high level will involve observing indicators and driving actuators. More specifically, Electrical BUS will interact with the PAYload, RF, and EPS subsystems to manage the state of affairs on the CubeSat. Operations of the electrical subsystem may include directing the PAYload subsystem to send data to the RF subsystem for downlinking, and acquiring uplink data from RF to perform a reboot.

Literature Review Methodology

As it pertains to the design, the main components that make up the electrical subsystem essentially boil down to three categories: control unit (that is, device(s) responsible for carrying out generic operations of the electrical subsystem, for example an MCU), protocol (that is, an interface for connecting two or more devices and allowing them to share data in one way or another, for example I2C), and peripherals (that is, device(s) responsible for a particular function within the Electrical BUS subsystem, for example a watchdog). This classification is not a recognized one, but has been found to be useful when conducting this literature review since the overwhelming majority of the designs can be deconstructed into components each of which would fit into one of these categories [subjective].

In terms of sources, two major categories are surveyed:

- Commercial OBCs discussed in NASA's Small-Sat State of the Art reports from 2022 to 2026 ([**?** , pp. 215–226], [**?** , pp. 225–237], [**?** , pp. 227–239], [**?** , pp. 241–253]), with trends also observed.
- OBCs built by academic design teams around Canada and beyond (considered designs listed in Appendix A)

Document Revisions

Version	Date	Author	Description
0.1	July 2026	Boris	Initial

Design Space

This section outlines the resulting design space for the Electrical BUS Subsystem based on the corresponding literature review. We establish a space of common control units, protocols and periptherals for our internal design process.

Control Units

The control-unit design space ranges from low-power microcontrollers to heterogeneous processor–FPGA systems. NASA's State of the Art reports show that most conventional CubeSat OBCs use microcontrollers or embedded microprocessors because command execution, telemetry collection, timekeeping, fault management, and subsystem coordination do not require high computational performance [**? ? ? ?**]. Higher-performance FPGAs, application processors, and system-on-chip devices are generally introduced when the OBC must also support image processing, autonomy, high-rate payload data, or machine-learning workloads.

Table 1: Candidate OBC control-unit architectures

Architecture	**Characteristics**	**NASA SOA Examples**	**Academic Examples**
Single MCU or processor	One device performs command execution, telemetry, timekeeping, fault management, and subsystem control. Lowest complexity and generally lowest power.	EnduroSat OBC, ARM Cortex-M7 [**?** , p. 229]; AAC Clyde Space Kryten-M3, Cortex-M3 [**?** , p. 228].	STM32L4 PC/104 OBC; Athena TMS570; ORCASat TMS570 [**? ? ?**].
Multicore SoC	Multiple cores share one device and memory system. Provides additional performance or lockstep capability, but does not remove the SoC as a single point of failure.	Argotec FERMI, dual-core LEON3FT [**?** , p. 229]; SPiN MA61C, dual-core GR712RC [**?** , p. 236]; Xiphos Q7S, dual-core Cortex-A9 [**?** , p. 238].	Athena and ORCASat use lockstep-capable ARM safety microcontrollers [**? ?**].
Dual redundant processors	Two processor devices provide cold, warm, or active redundancy. One processor may take control following a fault in the primary device.	Andes OBC, two Cortex-M7 processors with full cold redundancy [**?** , p. 230]; Bradford Binary OBC, dual Cortex-R5 with internal cold redundancy [**?** , p. 229]; Bradford Stellar OBC, dual PolarFire SoCs with internal cold redundancy [**?** , p. 230].	No directly equivalent two-device redundant architecture was identified in the academic survey.
Controller plus FPGA or DPU	A low-power controller performs critical spacecraft control while an FPGA or higher-performance processor handles payload data or acceleration. The secondary unit may be powered down when unused.	KP Labs Antelope, RM57 OBC plus Zynq UltraScale+ DPU [**?** , p. 230]; NOVI Gen 2, Vorago MCU plus Versal AI Edge SoC [**?** , p. 232].	NEUDOSE, primary AVR32 plus secondary Xilinx Zynq-7000 [**?**].
FPGA-centric	Control or processing is implemented using programmable logic and, optionally, a soft processor. Suitable for deterministic parallel interfaces and high-rate data paths, but more difficult to develop as a general-purpose OBC.	LASP Generic SmallSat FPGA Board, Kintex-7 [**?** , p. 230]; KP Labs Lion, Kintex UltraScale [**?** , p. 230].	No FPGA-only primary OBC was identified in the academic survey.

Table 1: Candidate OBC control-unit architectures (Continued)

Architecture	Characteristics	NASA SOA Examples	Academic Examples
Distributed controllers	Subsystem-local processors control the EPS, ADCS, communications, or payload and exchange data with a coordinating OBC. This improves fault containment but increases network and software complexity.	NASA identifies federated, distributed, heterogeneous, and mixed-criticality avionics as viable SmallSat architectures [? , p. 225] [? , p. 221–222].	Individual subsystem controllers used alongside the primary academic OBC.

All architectures will be considered during preliminary design; however, the current baseline is a centralized microcontroller-based OBC with independent watchdog and reset circuitry. This approach is the most common among the academic systems surveyed and provides the lowest-risk balance of power, determinism, software complexity, and flight heritage. An FPGA or secondary processor may be added later if payload-processing requirements cannot be met by the primary MCU.

Protocols and Interconnect

This section will summarize the survey or protocols across the NASA's SOAs as well as Canadian design teams' CubeSats in Table 2, as well as provide an overview of interconnect architectures in Table 3.

Table 2: Protocol and electrical-interface design space

Protocol	Representative Rate	Survey Examples	Primary Considerations
I^{2}C	100 kbit/s–3.4 Mbit/s	LORIS OBC [?]; identified as a common synchronous interface in all four NASA surveys [? , p. 219] [? , p. 235] [? , p. 240] [? , p. 224].	Low pin count and broad peripheral support; limited distance and bandwidth; shared-bus faults may block all devices.
SPI	Application-dependent; commonly 1–50+ Mbit/s	LORIS OBC [?]; NASA interface surveys [? , p. 219] [? , p. 235] [? , p. 240] [? , p. 224].	Deterministic and comparatively fast; requires separate chip-select signals and provides no standardized addressing or error handling.
UART / RS-422 / RS-485	Typically kbit/s to approximately 10 Mbit/s	LORIS uses UART [?]; differential serial interfaces are listed by NASA [? , p. 219] [? , p. 235] [? , p. 240] [? , p. 224].	Simple implementation. RS-422/485 provides greater noise immunity and cable length than logic-level UART; termination and driver control are required.
CAN / CAN FD	1 Mbit/s Classical CAN; commonly up to 8 Mbit/s CAN-FD data phase	STM32L4 PC/104 OBC [?]; CAN is listed as a common spacecraft fieldbus [? , p. 219] [? , p. 235] [? , p. 240] [? , pp. 224–225].	Differential multidrop operation, arbitration, acknowledgements, and error confinement make it suitable for EPS, ADCS, RF, and housekeeping traffic. Payload throughput is limited.
USB 2.0	Up to 480 Mbit/s	LORIS OBC [?].	High COTS compatibility and throughput, but host/device roles, connector retention, software complexity, and limited spacecraft-bus heritage must be considered.

Table 2: Protocol and electrical-interface design space (Continued)

Protocol	Representative Rate	Survey Examples	Primary Considerations
Ethernet	100 Mbit/s–1 Gbit/s commonly	Listed as a network interface in each NASA survey [? , p. 219] [? , p. 235] [? , p. 240] [? , p. 224].	High throughput and broad software support; requires switching or routing and may require additional measures for deterministic real-time operation.
SpaceWire / SpaceFibre	SpaceWire: typically 10–400 Mbit/s; SpaceFibre: multi-Gbit/s	SpaceWire is listed in the 2022–2024 surveys [? , p. 219] [? , p. 235] [? , p. 240]; SpaceWire and SpaceFibre are included together in 2026 [? , p. 225].	Space-qualified point-to-point networking with routing and fault-management support. Requires more specialized hardware and software than CAN or UART.
SerDes / MGT	Gigabit-per-second class; XAUI provides 10 Gbit/s aggregate	Identified by NASA as high-rate mission-data interfaces [? , p. 219] [? , p. 235] [? , p. 240] [? , p. 225].	Suitable for FPGA, payload, mass-memory, and radio data paths. Requires controlled-impedance routing, suitable connectors, reference clocks, and signal-integrity analysis.
GPIO / PWM / ADC / DAC	Application-dependent	LORIS provides GPIO, PWM, and ADC [?]; discrete and analog interfaces are listed by NASA [? , p. 219] [? , p. 235] [? , p. 240] [? , p. 225].	Appropriate for enables, inhibits, deployment signals, and analog telemetry; creates dedicated wiring and provides little inherent fault detection.
JTAG / ISP / ICSP	Not a mission-data interface	Debug and programming interfaces are listed in the NASA surveys [? , p. 219] [? , p. 235] [? , p. 240] [? , p. 225].	Required for programming and board-level debugging; access should be isolated or protected during flight.

Table 3: Candidate interconnect architectures

Architecture	Connection Pattern	Advantages	Limitations
Unified stack	OBC, EPS, RF, ADCS, and payload share a PC/104-style stack-through connector carrying power and common buses. The STM32L4 academic OBC uses this format [?]. NASA discusses PC/104 and stackable mezzanines in [? , p. 210], [? , pp. 226–227], [? , pp. 229–230], and [? , p. 222].	Compact, modular, and requires little harnessing.	Shared pinout constrains all boards; board order and connector bandwidth may become limiting.
Backplane	Subsystem cards plug into a shared CompactPCI or SpaceVPX backplane [? , p. 210] [? , p. 226] [? , p. 229] [? , p. 222].	Rigid mechanical support, modular replacement, and high-speed routing.	Higher mass, volume, connector cost, and chassis complexity.
Centralized point-to-point	Each subsystem has a dedicated connection to the OBC; all commands and data pass through the OBC. NASA identifies direct point-to-point and centralized architectures in [? , p. 207] and [? , p. 226].	Simple command authority and strong link-level isolation.	High connector count; OBC becomes a bandwidth bottleneck and single point of failure.

Table 3: Candidate interconnect architectures (Continued)

Architecture	**Connection Pattern**	**Advantages**	**Limitations**
Shared peer network	OBC, EPS, RF, ADCS, and payload controllers communicate as peers over CAN, Ethernet, or SpaceWire.	Reduces wiring, allows direct subsystem communication, and supports distributed control.	Requires addressing, arbitration, synchronization, and network-level fault management.
Payload pass-through	The OBC controls the payload, while high-rate data travels directly between the payload, mass memory, FPGA, and RF subsystem.	Avoids loading the housekeeping bus or copying all payload data through the primary MCU.	Requires separate control and data paths, flow control, buffering, and an OBC mechanism to isolate or reset the data path.
Hybrid dual-plane	A low-speed CAN or stack bus carries commands and housekeeping data, while dedicated Ethernet, SpaceWire, or SerDes links carry payload data. NASA describes this mixed topology in [**?** , pp. 222–223].	Combines a robust housekeeping network with scalable payload throughput and fault isolation.	Requires two interface classes and clear ownership of commands, telemetry, and high-rate data.

For the mission, the interconnect as well as the protocol bouquet selection will want to target higher throughput interfaces for RF and PAY subsystems, and thus will generally be dictated by them. In the spirit of a generic design, selecting a highet-speed interface is favoured, yet this might collide with the interconnect architecture selection since certain interfaces require specific connections that may not for example berouted through PC104 due to signal integrity requirements (e.g. SpaceFibre). The stack-up requirements and/or constraints might also be driven by Mechanical subsystem.

Peripherals

This section reviews the common peripherals that are found on OBCs surveyed by NASA's SOAs, as well as on OBCs designed by Canadian academic design teams. Details about volatile and non-volatile memory can be found in Tables 4 and 5, respectively. Other types of peripherals are summarized in Table 6. General considerations related to peripherals revolve around external vs internal peripherals since modern day SoCs contain many of the necessary peripherals on the same chip.

Volatile Working Memory

Volatile working memory, which loses functionality when the electrical power is disrupted, acts as high-speed temporary storage. For a CubeSat, it enables real-time software tasks. The primary consideration during implementation is between internal and external memory: internal memory (built within the main processor chip) reduces component count and PCB complexity; external memory provides greater capacity but introduces additional power and routing requirements.

Table 4: Volatile Memory Types, Metrics, Considerations, and Examples

Memory Type	**Description, Considerations, & Examples**	**Estimated Speed**	**Estimated Capacity**
SRAM (Static RAM)	Typically internal cache or memory within an SoC, used for low-latency code execution. Surveyed by NASA [**?** , p. 218] [**?** , p. 234] [**?** , p. 239] [**?** , p. 223].	10-25 ns access time [**?**]	4-32 Mb [**?**]
DRAM (Dynamic RAM)	High-density external volatile memory that uses single-transistor, single-capacitor cells. Surveyed across NASA missions [**?** , p. 218] [**?** , p. 234] [**?** , p. 239] [**?** , p. 223].	25 ns access time [**?**]	N/A

Table 4: Volatile Memory Types, Metrics, Considerations, and Examples (Continued)

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
SDRAM (Synchronous DRAM)	High-density external memory synchronized with the system clock; utilized externally by RADSAT-SK for general runtime tasks [**?**].	N/A	N/A
DDR3 (Double Data Rate 3)	High-speed third-generation SDRAM; Dalhousie University's LORIS uses 256 MB DDR3 [**?**].	13-15 ns [**?** , pp. 67-70]	4 GB [**?** , p.1]
ECC RAM (Error-Correcting)	Hardware-protected external memory that automatically detects and corrects single-bit errors; utilized by ORCASat [**?**].	N/A	N/A

Non-Volatile Program and Mass Memory

Non-volatile mass memory is persistent computer storage that retains stored data without requiring a continuous power supply, unlike volatile memory. It can store system boot code, flight software images, and long-term scientific data. Unlike the considerations for volatile memory, selection for non-volatile memory depends on many factors, including capacity, write endurance, retention, access speed, boot reliability, and susceptibility to radiation-induced corruption. Redundant boot images and error detection should be considered.

Table 5: Non-Volatile Memory Types, Metrics, Considerations, and Examples

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
MRAM (Magnetoresistive)	High-endurance, radiation-tolerant non-volatile memory that relies on magnetic states to store data;typically used as internal or external storage for persistent system state. Used by the C3S OBC [**?** , p. 229] and in redundant configurations on the Andes OBC [**?** , p. 232] [**?** , p. 230].	300 ns access time [**?** , p. 224]	
FRAM (Ferroelectric RAM)	Low-power, fast-write non-volatile memory (internal or external) that stores memory due to its ferroelectric properties; electric dipoles stay in place even without power [**?** , p. 229].	300 ns access time [**?** , p. 224]	
eMMC (embedded Multi-Media)	High-density external mass storage featuring an integrated flash controller; used by the C3S OBC [**?** , p. 229] and by LORIS (4 GB eMMC) [**?**].		
NAND Flash	Bulk, block-based external storage; deployed in redundant configurations on the Andes OBC [**?** , p. 232] [**?** , p. 230].		
Flash	External high-density program memory used for flight software storage and telemetry archives, as seen on Athena [**?**] and ORCASat [**?**].	50 ns read access [**?** , p. 224]	

Table 6: Common OBC peripheral classes

Peripheral Class	Examples	Primary Considerations
Watchdog, reset, and supervisor circuitry	The C3S OBC uses a multi-level watchdog [**?** , p. 229] [**?** , p. 232] [**?** , p. 230]; the Aerospacelab OBC includes a radiation-hardened supervisor MCU and watchdog [**?** , p. 232] [**?** , p. 229]. NASA discusses watchdogs and power cycling as fault-mitigation mechanisms [**?** , p. 220] [**?** , p. 236] [**?** , p. 241] [**?** , pp. 220–222].	An external watchdog is preferable for recovery from processor lock-up. Consider independent clocks, reset coverage, timeout configuration, communications watchdogs, and selective power cycling.
Timers and timekeeping	NASA identifies PLLs, capture/compare units, and time-processing units as common OBC peripherals [**?** , p. 219] [**?** , p. 235] [**?** , p. 240] [**?** , p. 224]. Timers are integrated within the STM32L4, TMS570, and i.MX6DL devices used by the surveyed academic OBCs [**? ? ? ?**].	Used for scheduling, timestamping, watchdog servicing, and waveform generation. Clock drift, oscillator redundancy, synchronization, and operation during processor resets must be considered.
Analog and discrete I/O	NASA lists GPIO, ADC, and DAC peripherals in each avionics survey [**?** , p. 219] [**?** , p. 235] [**?** , p. 240] [**?** , pp. 224–225]. LORIS exposes GPIO, PWM, and ADC interfaces [**? ?**].	Used for analog telemetry, enables, inhibits, deployment signals, and simple sensors. Consider voltage compatibility, input protection, ADC accuracy, safe reset states, and whether signals require isolation or redundancy.
Debug and programming interfaces	NASA lists JTAG, ISP, ICSP, BDM, BITP, and DB9 interfaces [**?** , p. 219] [**?** , p. 235] [**?** , p. 240] [**?** , p. 225].	Required for programming, boundary scan, manufacturing tests, and debugging. Flight hardware should provide controlled physical access and prevent unintended entry into debug or programming modes.
Optional integrated navigation peripherals	The EnduroSat OBC is available with integrated GNSS [**?** , p. 229] [**?** , p. 230].	Integration can reduce board count and harnessing, but may couple the OBC lifecycle to a specific receiver and complicate antenna placement, RF isolation, replacement, and independent fault recovery.

[TODO: survey each class of peripherals in more detail]

COTS (Commercial Off the Shelf) Design Alternatives

Description: Determine what COTS solutions exist if we cannot produce a certain part of the subsystem in house. Find a COTS option for each in house design.

Will inform trades later on for COTS versus in house risks.

This section outlines alternative off-the-shelf components that would in part or in full fulfil the operating requirements of the subsystem.

TODO

System Architecture and Budgets

In this section we concern ourselves with the following considerations: power draw, physical dimentions, radiation tolerance, and weight for the purposes of the overall design and integration with other teams.

Power draw

Based on the commercial onboard-computing systems surveyed in NASA's State of the Art reports [? ? ? ?], OBC power consumption can be divided into three broad categories according to platform complexity and computational capability. The reports do not use a consistent average-power metric; reported values are variously identified as idle, static, typical, active-mode, full-load, or peak consumption. Consequently, Table 7 preserves the operating-state terminology used by the manufacturers.

The 2022 edition identifies representative onboard-computing products but does not tabulate their power consumption [? , Table 8-1, pp. 214–217]. Quantitative power values are therefore drawn primarily from the 2023, 2024, and 2026 editions [? , Table 8-1, pp. 228–233] [? , Table 8-1, pp. 230–236] [? , Table 8-4, pp. 229–234].

Table 7: OBC power-consumption trends and reported operating states

Platform Complexity	Reported Power Profile	Design Characteristics	Example Designs
Low (Bare-Metal / Control)	**Minimum listed:** 0.05 W **Idle/steady:** approximately 0.2–0.5 W **Peak:** generally below 1 W	Simple command and data handling, housekeeping, telemetry, and fault management. These systems are normally based on low-power microcontrollers and execute bare-metal software or a small real-time operating system. Their power consumption is relatively insensitive to computational load.	Pumpkin PPM family: 0.05 W, operating state not specified [? , p. 231] [? , p. 235] [? , p. 233]. AAC Clyde Space Kryten-M3: 0.4 W [? , p. 228] [? , p. 230] [? , Table 8-4]. C3S Electronics OBC: 0.46 W in test mode [? , p. 229] [? , p. 232] [? , p. 230]. SkyLabs NANOobc-2: $<$ 0.9 W idle and $<$ 1 W peak [? , p. 232] [? , p. 236].
Medium (Integrated OBC)	**Idle:** approximately 0.2–1 W **Nominal/steady:** approximately 1–3 W **Peak:** approximately 1.5–5 W	Typical integrated CubeSat OBCs combining command and data handling, mass-memory management, communication interfaces, fault protection, and basic payload management. Power may vary with processor utilization, interface activity, and memory access.	SPiN MA61C CubeSat: 1–1.2 W [? , p. 232] [? , p. 236] [? , Table 8-4]. AAC Clyde Space Sirius OBC & TCM: 1.3 W [? , p. 228] [? , p. 230] [? , Table 8-4]. EnduroSat OBC I: 0.2 W idle and 1.5 W peak [? , p. 229] [? , Table 8-1] [? , p. 230]. Resilient Computing RadPC: 1.5 W [? , p. 232] [? , p. 236] [? , p. 233]. SatRev OBC: 2.8 W typical and 5 W peak [? , p. 233].
High (Edge / AI / SoM)	**Idle/static:** approximately 5–15 W **Typical/active:** approximately 6–10 W **Peak/full load:** commonly 10–20 W, with larger payload computers reaching substantially higher values	Heterogeneous systems-on-module and payload-processing computers supporting image processing, software-defined radio, autonomous navigation, data compression, and machine-learning inference. Power is strongly dependent on processor utilization, FPGA configuration, accelerator activity, and payload duty cycle.	Novo Space GPU002AF: 6–11 W in active mode [? , p. 231] [? , p. 235] [? , p. 233]. KP Labs Leopard DPU: 7 W static and up to 10 W during deep-learning inference [? , p. 230] [? , p. 234] [? , p. 231]. Aitech S-A1760: $\leq$ 5 W idle, 8–10 W under a typical CUDA load, and $<$ 20 W at full utilization [? , p. 228] [? , Table 8-1] [? , Table 8-4]. EnduroSat GPC: $<$ 15 W idle and 130 W peak [? , p. 231].

Based on these results, a nominal power-design envelope of 1–10 W is adopted for the proposed OBC. This range encompasses conventional integrated CubeSat OBCs and moderate edge-processing workloads, while transient loads above 10 W are treated separately as peak operating conditions. The selected range is a project design boundary rather than the full commercial-hardware envelope, which extends from approximately 0.05 W for low-power microcontroller boards to 130 W peak for high-performance payload computers.

[TODO: peak loads and durations]

Dimentions

Weight

Radiation tolerance

Conclusion

This report outlined the state-of-the-art OBC review from both internal design stadpoint through the Design Space section as well as system integration standpoint through System Architecture section. Internal design is broken into selection of control unit, protocol set and interconnect topology, as well as periptheral array. Each of these compoenents have a wide selection of compoenetns and designs which will need to be researched further and used as part of trades once requirements are established. Further research and collaboration with other subsystems is also required to guide the internal design decisions for this subsystem.

Appendix A: Academic OBCs

Name	**Team**	**MCU**	**Protocols**	**Standard**	**Notes**	**Ref**
STM32L4 PC/104 academic OBC	VST104 thesis design	STM32L4—ARM-based	CAN	PC104	pc 104 cubesat design.	[?]
Athena OBC	University of Alberta (Albertasat)	TI TMS570LC4357 (ARM Cortex R5F)			Embedded memory type: Flash. Features ideal for satellite OBC application, including radiation & space reliability (rad-hard, SEL immune 48 MeV at 125°C, LAT 30krad), high computational performance (lockstep dual-core compare, 300MHz CPU clock, 1.66DMIPS/MHz).	[? ?]
Orcasat OBC	University of Victoria	TI TMS570LS0714 (Dual ARM Cortex-R4F)			Embedded memory type: Flash. Operates in lockstep, ECC integrated flash & RAM banks, radiation-tested up to 6 krad, 80MHz core clock.	[? ?]
LORIS OBC	Dalhousie University	NXP i.MX6DL (Single-core ARM Cortex-A9)	USB 2.0; SPI; I2C; PWM; UART; GPIO; ADC		Embedded memory type: Flash. Low profile, power efficient SoC SBC. 4GB eMMC flash storage, 256MB DDR3 RAM, 800MHz operational frequency, -40°C to 85°C temperature range.	[? ?]
NEUDOSE OBC	McMaster University	Primary: AVR32. Secondary: Xilinx Zynq-7000 SoC			Primary is COTS chosen for high TRL to reduce mission risk; secondary is designed for radiation tolerance using rad-hard and non-rad-hard components.	[?]
RADSAT-SK OBC	University of Saskatchewan	ARM926EJ-S			Embedded memory type: SDRAM. Uses ISISpace satellite-subsystems libraries, contains MMU, designed to interface with external SDRAM.	[?]